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CSTrainer : An Ontology Driven Multiple AIGC Agentic Approach for Oil and Gas Scientific Research Supporting

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Abstract

Scientific research in the energy and oil & gas sector faces significant challenges, including massive heterogeneous data, complex mechanisms, high uncertainty, and substantial reliance on expert experience. While recent advancements in generative AI offer autonomy in multi-step reasoning, generic models offered limitations such as hallucinations, insufficient domain knowledge, and poor alignment with researchers' prior works. To address the need for continuity in research trajectories, this study proposes CSTrainer, an ontology-driven, multi-agent system designed for energy-specific scientific research assistance. CSTrainer integrates 5 foundational models (GPT-4o, DeepSeek, Qwen, Tencent-Hunyuan, and Doubao-Seed (five foundation LLMs)) and employs a hierarchical architecture featuring multi-agent clusters, cross-model voting, and debate mechanisms. A core innovation is its use of a distributed dynamic ontology framework, which constrains reasoning paths to ensure scientific rigour and mitigate hallucinations. The system facilitates end-to-end research support, encompassing scientific problem formulation, technical route planning, ontology construction, and executable workflow generation, augmented by weighted Bayesian updating and lightweight human-in-the-loop (HIL) feedback. The system was validated on 3 datasets: UNISIM-IV (open), tight gas well re-evaluation (proprietary), and transnational pipeline scheduling (proprietary). Results demonstrate that CSTrainer achieves stable, high quality performance for 3–5 scientific questions extraction, outperforming agent-only baselines by 62.5%–87.5%. While 10 rounds of detailed human feedback yielded optimal results, excessive interaction or scaling beyond 10 problems induced instability. The system attained full marks in the tight gas re-evaluation scenario, confirming its efficacy in bridging the gap between conceptual research and engineering implementation. CSTrainer provides a scalable, interpretable, and controllable AI-for-science pathway for industrial research.

Keywords: Ontology; Scientific research supporting; Energy engineering research; Multi-Agent; Generative AI

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Introduction

Scientific research in the energy industry faces multiple challenges, including massive and heterogeneous data, complex underlying mechanisms, high costs of trial-and-error, and a heavy reliance on expert experience (Huang et al. 2025). The conventional research paradigm typically commences with extensive literature surveys and researchers' personal insights into domain-specific problems. This process, which relies heavily on deep manual analysis and experts' interpretation, is often inefficient and constrained by the cognitive boundaries of individual researchers. Recently, a series of multimodal AIGC models, such as GPT and DeepSeek etc, have demonstrated formidable capabilities in knowledge comprehension, knowledge integration, hypothesis generation, and logical reasoning (DeepSeek 2026; OpenAI 2026), offering new possibilities for building intelligent research assistants (Yuan et al. 2025). However, the direct application of general-purpose foundational models (LLMs) to specialised fields presents dual challenges. On one hand, they suffer from inherent limitations of AIGC, such as "hallucinations", insufficient depth in domain-specific knowledge, uncontrollable reasoning processes, and difficulty in generating structured, executable solutions. On the other hand, limitations in in-context learning hinder these models from fully comprehending a user's prior research foundation and preferences to generate extended research directions (Thirumaran et al. 2013). Therefore, exploring methods to integrate the capabilities of LLMs with domain expertise, controllable workflows, and users' research history to develop specialised research assistance tools for oil and gas industry domain has become a key to implementing AI for science (AI4S) in industrial settings.

Related works

Represented by models like AlphaFold, AI4S has achieved a leap from "data-driven" to "knowledge-driven" exploration through sequence prediction, structure simulation, and system architecture reconstruction. This has established a "sequence-structure-inference" technical pathway for protein prediction. Its technological innovation and application expansion have not only reshaped the research paradigm in structural biology and advanced the deep integration of AI in interdisciplinary scenarios such as antibody and enzyme design, drug discovery, and immunotherapy, but also furnished a complete technical pipeline encompassing "accurate prediction, quantitative analysis, and application translation" for AI-driven research (Xu et al. 2025; Fan et al. 2025). The interdisciplinary research potential unlocked by such AI also presents a novel technical route to address the innovation bottlenecks stemming from single-discipline silos in current scientific research (Kargupta et al. 2026; Ding et al. 2025). Specifically, Idea-Catalyst proposes a novel framework (Kargupta et al. 2026) centered on meta-cognition, which effectively addresses the traditional scarcity and superficiality of interdisciplinary research through a three-stage process of problem decomposition, cross-domain exploration, and insight reconstruction. This framework enhances the novelty and depth of generated ideas by 21.38% and 16.22%, respectively, providing a structured tool for AI-assisted brainstorming (Ding et al. 2025). Meanwhile, a bibliometric analysis reveals the interdisciplinary diffusion characteristics of GenAI from computer science to fields like medicine and sociology, confirming that while boosting research productivity, it has fostered a unique collaborative model featuring "smaller team sizes yet higher international collaboration" (Ding et al. 2025). However, on one hand, the output of the Idea-Catalyst framework suffers from redundancy, lacking conciseness and operationalizability in its interdisciplinary insights. Although GenAI-related research shows a cross-domain diffusion trend, its reliability in high-precision scientific scenarios is still questionable due to limitations in data integrity and model "hallucinations", and a deep exploration of the differentiated needs and adaptation pathways for GenAI across disciplines is lacking. Regarding the robustness of AI-assisted scientific computing, empirical analyses across five core scientific domains, including weather prediction, chemical energy calculation, and fluid dynamics, reveal the high sensitivity of neural networks to subtle perturbations, including FGSM adversarial attacks and random noise (Zhang et al. 2024). This inherent vulnerability

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can cause model outputs to deviate significantly from true values, leading to potential risks like erroneous scientific conclusions and flawed engineering decisions. It has also been shown that introducing external input to smooth such fluctuations can help mitigate the hallucination problems caused by these disturbances (Zhang et al. 2024). While these studies collectively demonstrate the significant efficiency gains GenAI brings to the research domain, they also further substantiate that AI4S, as a typical HIL scenario, faces challenges: insufficient robustness to minor perturbations affecting research reliability; poor cross-context and human computer interaction capabilities, making it difficult to achieve reasonable, research based ideation divergence from a user's existing foundation; and a lack of interpretability and reliability in multi-step, time-varying scientific scenarios, where models are prone to overfitting or generating hallucinations due to excessive divergence.

The emergence of autonomous intelligent agents (AI Agents), exemplified by systems like OpenClaw, signifies AI's evolution from a passive tool to an active partner capable of planning, collaboration, and "autonomous" research and evolution (Dang et al. 2025; Hu et al. 2025; Lyu et al. 2026). For AI4S scenarios, which inherently rely on intensive HIL interaction and require rich imagination to better handle uncertainty, the paradigm of GenAI+Agent is evolving from single-agent operation towards multi-agent, knowledge and data co-driving models. This evolution is further integrated with generative digital twin simulation technologies to enable better scientific problem discovery and solving. This technological breakthrough partially addresses the bottlenecks of using AIGC alone for research activities. By constructing complex human-machine collaboration and multi-agent systems, AI is beginning to deeply intervene in and restructure the entire research cycle of "hypothesis formulation, experimental design, and verification analysis." In prior research on GenAI agent-based AI4S, the ChatBCI toolbox, based on a dual-sided design principle, facilitates efficient collaboration between human experts and AI in Brain-Computer Interface (BCI) research through flexible autonomy and a shared knowledge base, addressing pain points like scarce domain data and complex signal analysis (Kapitonova&Ball 2024). For data-intensive sciences like high-energy physics, a framework based on "human strategic guidance + agent technical execution" was developed, employing a dedicated domain language (SaiScript) as a human-machine interaction board This framework losslessly translates human research intent into executable analytical workflows, breaking through traditional efficiency bottlenecks and providing a transferable human-machine collaboration method for AI-assisted handling of complex problems in high-energy physics (Li et al. 2026).

To navigate the complexity and uncertainty of research activities and maximise the depth of collaboration between human expertise and genAI, researchers have innovated various multi-agent system architectures. The core idea is to simulate and optimise the operation of a research ecosystem through agent specialisation, human-machine collaboration, and competitive mechanisms. These innovations primarily encompass three categories. The first involves building self-contained loops to assist researchers in generating research outputs. For instance, the Dolphin framework automates the iterative process from literature analysis to code debugging through agents driving idea generation, experimental validation, and result feedback (Yuan et al. 2025). EvoScientist constructs a self-evolving research multi-agent system (Lyu et al. 2026), where the division of labour among researcher, engineer, and evolution manager agents, combined with dual memory modules for ideas and experiments, enables full-process experience accumulation and iterative optimisation. A couple of paper automatically generated by this system was accepted at conference and won an award, validating the architecture's practical value in producing original outcomes. A second line of research introduces a mechanism design perspective to address autonomy bottlenecks in research assistance systems. MACC models the multi-agent system not as a single entity but as a "simulated scientific community" composed of independent agents (Oyama et al. 2026). By designing mechanisms like a shared workspace, reward information sharing, and result reproducibility verification, it establishes a "collaboration-competition" balance among agents, systematically addressing institutional challenges in traditional research such as exploration redundancy and low reproducibility, thus providing a scalable theoretical and experimental platform for large-scale distributed scientific exploration. Concurrently,

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addressing the common issues of continuous task definition and reward shaping in AI-driven research tasks. From user intent direction, the MTIRL system explores a method of human intent driven multi-agent reinforcement learning. It aggregates feedback from multiple human trainers and establishes a dynamic trust model to convert dispersed human experience into reliable reward signals, mitigating the bias and limitations of existing single feedback approaches and offering a novel agent-to-agent training paradigm for research tasks with sparse or ill-defined rewards (Guo et al. 2022).

A third category deeply aligns with industrial application needs, advancing the engineering application of AI research in fields like chemistry and materials science. In chemical engineering, to bridge the "from AI discovery to industrial production" gap, AutoChemSchematic proposes a closed-loop, physics-aware automation system (Srinivas et al. 2025). It integrates domain-specific small language models, chemical engineering knowledge graphs, and process simulators to automatically generate and simulate process flow diagrams (PFDs) and piping & instrumentation Diagrams (PIDs) from molecular designs. Through multi-stage fine-tuning, GraphRAG, and simulator-in-the-loop verification, it ensures the feasibility and economic viability of process designs, providing an end-to-end solution for AI deployment in pharmaceuticals, chemicals, and related industries. In materials science, diffusion-based AI models like MatterGen are accelerating the discovery of novel high performance functional materials (Zeni et al. 2025). This model, by tailoring the diffusion process to the periodic structures of functional materials and incorporating property oriented adapter modules, can stably generate new materials that simultaneously satisfy multiple property constraints. Its yield of stable, unique, and novel materials significantly surpasses that of traditional methods, providing a powerful generative engine for material innovation in sustainable technologies like carbon capture and energy storage.

Although prior experiments have confirmed that GenAI driven multi-agent allied frameworks can provide significant assistance for scientific research activities, inherent limitations persist. Agents and genAI struggle to comprehend data lacking semantic content or with ambiguous semantics (Mai et al. 2024). Furthermore, the task chains planned by agents possess a degree of inherent divergence, which impairs reproducibility, the ability to build upon a researcher's existing foundation, and the accuracy, diversity, and reliability of scientific exploration. This introduces risks and uncertainties into AI-driven scientific research (Shimomura et al. 2025). Ontology, as a method of normalised semantic representation, provides a reliable approach for integrating and consistently representing complex, multi-source, heterogeneous, high-dimensional, and time-varying data (Chandra et al. 2025). It is now widely applied in complex decision making scenarios and explainable AI (XAI) applications. In constructing foundational research data infrastructure, the science datalake (Wilinski 2026) tackles issues of isolated academic databases, heterogeneous metadata, and a lack of mapping between subjects and ontologies. It achieves unified querying and high-precision semantic association across 293 million papers through multi-source academic data fusion, DOI normalisation, and a BGE embedding based ontology alignment method. This demonstrates the potential of an ontology based unified data foundation for supporting reproducible and scalable scientific research. Research has also methodically delineated the three-dimensional role of ontology in reference modelling, commonsense reasoning, and knowledge refinement/complexity management. This approach addresses problems in existing XAI, such as a lack of commonsense grounding, explanations misaligned with human understanding, and unmanageable complexity, thereby effectively enhancing the trustworthiness and explainability of AI decisions (Confalonieri&Guizzardi 2023).

Domain specific applications further illustrate its utility. In intelligent transportation, a method combining a road traffic domain ontology with graph-matching filters and a joint learning approach for vision-language models (VLMs) and diffusion models has been developed. This enables proactive inference of accident causes and the generation of credible improvement solutions, significantly enhancing the convergence efficiency of autonomous driving systems and improving environmental safety and solution explainability. In mathematical research, the integration of the OpenMath mathematical ontology with hybrid retrieval, cross-encoder re-ranking, and neuro-symbolic reasoning enhancement addresses issues like hallucinations,

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fragile reasoning, and a lack of formal grounding for LLMs in mathematics. It reveals the constraining effects of retrieval relevance, model capacity, and task difficulty on enhancement efficacy, achieving bidirectional improvement in both the quality and efficiency of mathematical problem retrieval (Labre 2026). For human-machine collaboration in industrial manufacturing, an architecture integrating NLP, ontology/knowledge graphs, and LLMs has been proposed. This knowledge enhanced LLM framework effectively tackles challenges of low accuracy, poor semantic fusion, computational imbalance, and opacity in industrial human-machine collaboration, providing a viable path for precise, industry-specific collaborative decision-making.

Building on this foundation, this manuscript proposes an ontology enforced multi-agent system for scientific research assistance in the energy and oil & gas industry, namely, CSTrainer. The system integrates multiple foundational LLMs, a group of collaborative agent cluster, and ontological modelling capabilities. It inherits a user's prior research outputs, employs collective intelligence to associate related studies, and leverages a distributed ontology framework to achieve digital twin like refinement of scientific questions via HIL confirmation, followed by autonomous generation of technical roadmaps and executable research plans. The methodology is validated on both public and proprietary datasets from upstream exploration, development and downstream pipeline operation decision making issues. Using a combination of objective metrics and subjective expert evaluation, the validation confirms CSTrainer's end-to-end research assistance capabilities, from automated problem formulation, technical planning, to executable workflow generation, as well as the problem-sorting regarding to each specific task. The remainder of this manuscript is structured as follows: Section 3 details the construction and core operational mechanisms of the CSTrainer system, covering the system architecture, scientific problem elicitation operations, foundational ontology framework, and detailed workflow modelling. Section 4validates the effectiveness and scenario generalisation capability of CSTrainer using three distinct case studies: reservoir analysis research, well-log interpretation research, and trans-national pipeline decision supporting. Summary and discussion of the results are given in section 5.

CSTrainer system

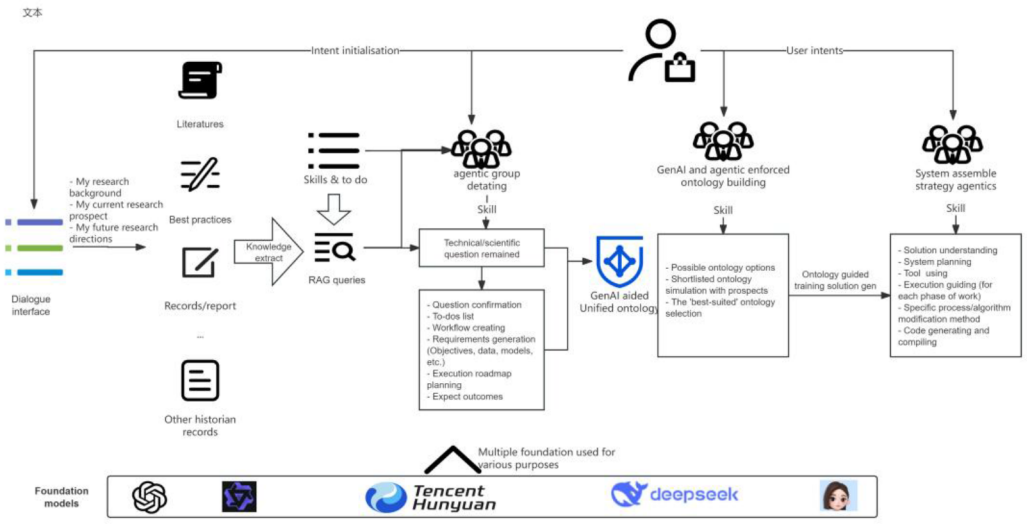
CSTrainer is designed to assist researchers by integrating multiple open source foundational models, skill templates, and agent clusters. The system facilitates the extraction of scientific questions suitable for the next research phase from historical research outputs, identifies potential issues, enables collaborative planning, and generates workflows alongside training and inference schemes for sub-task models. The overall execution process is illustrated in fig—1(a). The system integrates five commonly used foundation models, namely, GPT-4o, Tencent Hunyuan (TH), DeepSeek-V4Lite (DS), Qwen-3.5 (QW), and Doubao-Seed 7.0 (DB) to support the aforementioned research assistance tasks. At the outset, a researcher creates a new research task via a conversational interface, provides a task description, submits prior research outputs deemed relevant, and indicates preferences for future work. The system then employs a multi-model ensemble to formulate a unified skill, with a structured to-do list for the specified requirement.

Guided by the logic and requirements of the formulated skill, the system autonomously logs into institutional Athens accounts to retrieve relevant literature resources. It subsequently proposes scientific questions. A multi-agent debate mechanism, augmented by HIL evaluation, is employed to select candidate scientific questions that align with the researcher's intent and preference. For each specific scientific or technical problem, a diffusion and voting mechanism deduces and selects the most suitable ontology components. These components are combined to form a cohesive ontology, which serves as the problem solving blueprint.

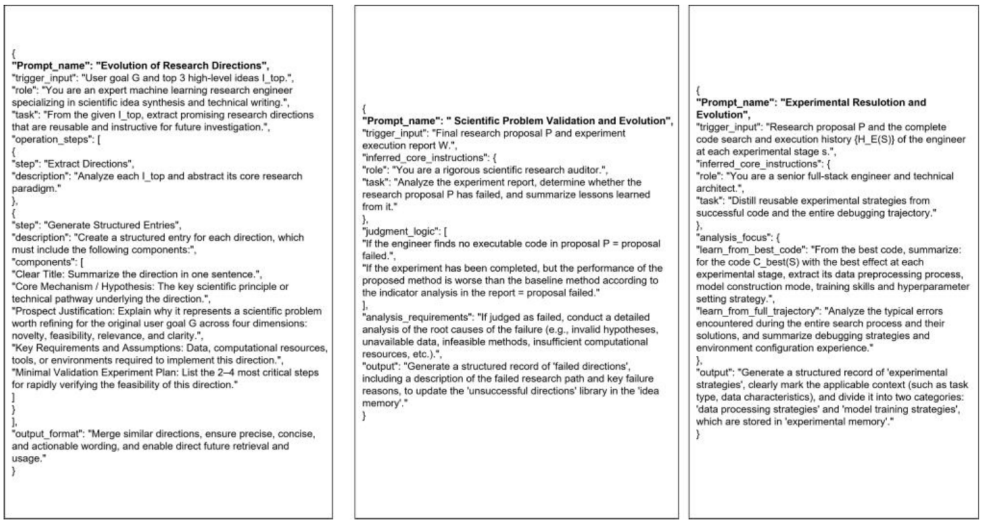
Subsequently, the problem solving agents interprets the execution plan based on this ontology, performing system planning, workflow generation, and system process modelling, and can even progress to specific workflow execution. As shown in fig—1, the CSTrainer's operation involves multiple rounds of human

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computer interactions and a decision-making/execution process based on collaborative debate and voting within the multi-agent cluster. Its essence lies in transforming a one-off interaction history (ideas, code, experimental results) into a structured, analysable, and continuously iterative research assistance activity.



(a)



(b)

Figure 1—(a) System architecture of CSTrainer; (b) Instance of evolutionary scientific research skill

To ensure the agent cluster evolves in a specified direction throughout this process, the study designed three specialised prompts, corresponding to the architecture of CSTrainer: 1) a research direction evolution skill for scientific problem elicitation, 2) a solution verification evolution skill for technical roadmap planning and verification, and 3) an experimental strategy evolution prompt for execution plan validation. These prompts guide the whole execution workflow of agent cluster during 3 core stages, as depicted in fig 1(b).

Main execution procedure and agent skill fitting by CSTrainer

At the initial stage of a task, the user inputs their research intents, termed the future works, via the human-computer interaction interface. The user also uploads their prior research knowledge and contributions (which can be imported via platforms like web of science/ORCID or through offline files) and authorises

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CSTrainer to utilise their institutional Athens account to search for related academic literature. Integrating the user's prior research with relevant findings from the broader research community, CSTrainer employs an auto generated skill template to structure the entire task planning process. First, guided by the user's intent, GPT determines the underlying logic for eliciting scientific/technical problems, which is then confirmed via a HIL verification step. Based on the confirmed problem-elicitation logic and the user's prior research contributions, the other 4 foundation models, i.e. TH, DS, QW, as well as DB are used to generate N candidate solutions for each procedural step (where N is user-defined). A cross voting mechanism among these 4 models is then applied to establish a global ranking of the proposed solutions. The highest ranked method is selected as the selected step. In scenarios where a tie occurs among the candidate options from the TH, DS, QW, and DB models, GPT acts as a weighted arbiter. Leveraging its comprehension of the user's original intent, it breaks the tie by selecting the final candidate from the tied set. The skill for addressing the current problem is subsequently generated by GPT-4o, which combines the selected steps that chosen from the candidate pools, guided by the user's future works intent. The mathematical representation for the operations described above is detailed in Tab 1.

Table 1—Algorithm of research procedure and skill fitting by CSTrainer

Algorithm1: research procedure and skill fitting
1: **Input:** User U, User Intent I (e.g., "future works"), User's Past Research P
2: **Output:** SelectedSolution (The highest-ranked candidate solution)
3: **Procedure:**
4: // Step 1: User Initialisation
5: Authorise U and upload P; Input I.
6: // Step 2: Define Problem Refinement Steps with Human Confirmation
7: Generate potential refinement steps S_GPT using GPT-4O based on I.
8: Obtain final steps S after user confirmation via Human In Loop Confirmation(S_GPT).
9: // Step 3: Multi-Model Candidate Generation
10: For each step s_i in S:
11: For each base model M_j in {TH, DS, QW, DB}:
12: Generate N candidate solutions: Candidates(i,j) = M_j(P, I, s_i).
13: // Step 4: Cross-Model Voting for Global Ranking
14: For each candidate c across all models for a given step:
15: Aggregate vote scores from all models M_j.Vote(c, candidate_set).
16: Rank all candidates globally by total vote score (RankedCandidates).
17: // Step 5: Select Top Solution with GPT Tie-Breaker
18: Identify top candidates (TopCandidates) with the highest vote.
19: **if** multiple top candidates exist, use GPT-4O (weighted by I) to break tie and select
20: final SelectedSolution.
21: **Return** SelectedSolution.

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Scientific and technical question extraction and optimisation

In terms of scientific problem formulation, CSTrainer builds upon the problem solving approach established in the preceding stage. Employing a core technical pathway centred on multi-agent collaborative reasoning, RAG, and iterative human-computer interaction, it constructs an end-to-end scientific problem formulation framework encompassing user intent parsing, knowledge fusion, problem divergence, intelligent screening, and continuous refinery. Leveraging the user's prior research contributions as a prior, the framework integrates domain knowledge with autonomous agent decision-making to achieve directed generation and dynamic fine-tuning of scientific questions. The specific process for scientific problem elicitation can be divided into the following key phases:

User intent initialisation and multi-source knowledge fusion. The system first facilitates the structured input of the user's research intent via the human-computer interface. This intent encompasses 3 core elements: research background, the current research project, and future research prospects. The user also defines the desired number of scientific questions to be generated, denoted as N (N, where N≥1). Concurrently, a multi-source, heterogeneous knowledge base is constructed, integrating resources such as research best practices, historical project records/reports, interdisciplinary references, and cutting-edge domain literature. Using natural language based RAG as the interaction medium, the system maps domain knowledge to the user's personalised requirements.

Decomposition and task-oriented modelling. Based on the user input skill, the system atomically decomposes the research task, generating technical or scientific question, as well as clarifies the core scientific contradictions and subtasks to be addressed within the research direction. It transforms abstract skill demands into executable reasoning tasks (including objective decomposition, requirement elicitation, and execution roadmap planning), providing task-oriented constraints for the multi-model, multi-agent problem formulation process. This ensures a strong correlation between the generated problems and the research goals.

Autonomous divergence mechanism based on multi-agent collaboration. 5 distinct agent clusters, each endowed with supervision-execution-review capabilities (fig—1), are designed. Each cluster is paired with one of the base models mentioned earlier. Drawing on established design paradigms for multi-agent clusters in long sequence or multiple steps scenarios within the oil and gas domain (Chen&Jin et al. 2025; Chen&Yang et al. 2025), each cluster contains at least: 1) a planner responsible for planning the executive chain and technical pathway for problem formulation, and determining the debating mechanism; 2) a supervisory agent that monitors the rationale of agent-generated content, activates the base model's deep thinking capability directionally for timely correction; and 3) multiple executor agents (for diverging and formulating scientific questions from different angles) along with corresponding debating agents. The number of executor agents required for a given problem formulation intent is autonomously determined by the supervisory agent paired with each base model.

At the start, each agent cluster, guided by the skill's predefined intent, autonomously determines the execution strategy for the problem formulation task (e.g., theoretical deduction, experimental validation, data-driven, interdisciplinary integration, application scenario oriented) and plans the cluster's structure. It then independently generates scientific questions by diverging from the “user intent + skills”. During problem formulation, an attention spreading mechanism (Chen&Zhao et al. 2025) is introduced within the agent clusters. Its core function is to spread the event association during scientific problem exploration, based on the reasoning task and dimensions such as the user's inferred future research preferences, related literature, and key unresolved industry issues, thereby activating executor agents to traverse more possibilities to mine candidate scientific questions, refers to as $N_{Candidate}$, where, $N_{Candidate}>N_{Selected.}$

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Scientific problem selection via multi-agent debate. Multi-agents debate mechanism assists the user in selecting the most reasonable scientific questions through a combination of internal and external debate. After all 5 agent clusters complete their initial problem formulation, an inner debating phase calls within each cluster. Here, debating and executor agents engage in autonomous debate, ranking the problems based on the user's prior research background and related literature knowledge. They construct an "assertion-rebuttal-consensus convergence" argument matrix across 3 dimensions: problem innovativeness, feasibility, and alignment with research objectives, resulting in a ranked set of candidate scientific questions: $N_{Candidate} = \{N_{GPT}, N_{DS}, N_{TH}, N_{QW}, N_{DB}\}$, where, the 5 variables correspond to the candidate sets generated by the 5 respective foundation models and agent clusters.

The candidate sets selected via inner debating are then subjected to cross voting by 5 supervisory agents serving by 5 foundation models. This vote selects the top N candidate scientific research problems, forming the set $N_{nt} = \{Q_1, Q_2, \ldots Q_N\}$. The voting process employs evaluation methods (e.g., Bayesian probability, Markov decision process, information entropy) autonomously chosen by each supervisory agent as it deems most appropriate, without human guidance or intervention.

Iterative optimisation and solution finalisation. Given the subjective nature of scientific questions and the complexity of research contexts, which directly impact subsequent workflow generation and technical planning, the final set N_{nt} requires user confirmation. If any problems are determined infeasible or low confidence by the user, CSTrainer automatically extracts the points of disagreement and the user's improvement prompts. Based on the user's specified requirements, it reverts to either the "multi-model divergence" or "agent debate" stage for re-execution. This loop continues until the generated problem set achieves a balance between innovativeness, feasibility, and user intent.

Since problem formulation must align with the user's further research interests and expected direction, multiple valid answers are possible. This phase still supports HIL intervention. If the agent debate outcome significantly deviates from user expectations, CSTrainer allows user to inject corrective instructions (prompts) via the interface. This guides the relevant agents to dynamically adjust their selection logic, preventing the generation of unexpected outcomes to the research direction, as illustrates in Tab 2.

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Table 2—Scientific question generation and refinery

Algorithm 2: Adaptive multi-role agents based scientific question generation & refinery

Input: U_bg (background), U_cp(Current prospect), U_fd (Future direction),N, S_user, KB

Output: Q∗ (final N scientific questions)

1: U_intent←StructuringModule(U_bg, U_cp, U_fd)

2: C_rag←Retrieve (KB,FormulateQuery(U_intent, S_user))

3: T_residual←AtomicDecomposition(S_user), Task_Set←{ObjDecomp,ReqElicit,...}

4: BaseModels←{M|GPT,DS,,TH, QW, DB}, Roles←{Supervisor, Planner, Executor...}, ...

Candidate_Pool←∅

5: for

6: M∈ BaseModels

7: do

8: Cluster_M←InitialiseCluster(M,Roles)

9: Strategy_M←Planner_AutoDesign(Cluster_M, S_user,U_intent)

10: Candidates_M←∅

11: for

12: T∈ Task_Set

13: do

14: for

15: E∈ (Cluster_M, Executors)

16: do

17: q_{cand}←E.Generate(T,C_rag,U_intent,T_residual) // To generate candidate questions

18: Candidates_M←Candidates_M∪ {q_{cand}}

19: end for

20: Candidates_M←Candidates_M∪Diffusion(Candidates_M, C_rag, Strategy_M)

21: end for

22: Candidate_Pool←Candidate_Pool∪ {Cluster_M.Supervisor.Validate(Candidates_M)}

23: end for

24: Ranked_Pool←∅ // To form the candidate pool for ranking

25: for C_SetM∈ Candidate_Pool do

26: Ranked_SetM←InnerDebate(Cluster_M.Debators, Cluster_M.Executors,...

27: ConstructMatrix(C_SetM, U_intent, C_rag))

28: Ranked_Pool←Ranked_Pool∪ {Ranked_SetM}

29: end for

30: Vote_Count←∅

31: for Cluster_M do // Vote for Top N research questions

32: Top_NM←Cluste_rM.Supervisor.Evaluate(Ranked_Pool,N,Cluster_M...

33: Supervisor.ChooseMetric()) //Supervisor agent to form

34: for Q∈ Top_NM do

35: Vote_Count[Q]←Vote_Count[Q]+1

36: end for

37: end for

38: $Q_{candidate}$←SelectTopNByVote(Vote_Count,N)

39: repeat // Return final candidate question by supervisor agent

40: Feedback←GetUserFeedback($Q_{candidate}$) // Refinery by user feedback

41: if Feedback.confidence≥threshold then Q∗ ←$Q_{candidate}$, Break;

42: else if

43: Feedback.rollback_to=="divergence" then

44: Crag←UpdateContext(C_rag, Feedback.prompts); goto 4

45: else if Feedback.rollback_to=="debate" then

46: ClusterM.AdjustLogic(Feedback.instruction); goto 19

47: until Q∗ = ∅

48: return Q∗

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Establishment of ontology backbone

Following the completion of scientific problem formulation, CSTrainer proceeds to generate a corresponding ontology backbone based on the selected technical problems. This ontology is subsequently deployed to plan technical roadmaps, formulate workflows, and generate specific training or configuration schemes for the algorithmic models involved in each workflow module. This ensures that subsequent scientific research is executed more efficiently and autonomously within a controlled cognitive framework.

Given that ontologies targeting process industries comprise a synthesis of static entities, dynamic events, and activities (Arabi et al. 2025), and that scientific problem solving typically involves multiple technical pathways, the ontology constructed in this study exhibits relatively fixed static components (entities and classes). Conversely, the dynamic components—specifically events (phenomena), relations (computational attributes), and activities (functions)—are characterised by non-unique solutions. To address this, the study introduces a simulated a "decision-making conference mechanism". By simulating the generation of distributed ontologies, the agent cluster is encouraged to explore a wider array of feasible technical routes, thereby increasing the probability of obtaining optimal solutions.

During the ontology construction phase, CSTrainer accesses the 5 selected foundational models and generates 5 distinct researcher agent groups, aligned with the formulated scientific questions and input skills. Each group comprises a number of researcher agents, denoted as R(a)={R1, R2,…, RM}, where M is user-defined maximum number of researcher agents and must be an odd number. Each agent belonging to R(a) is randomly assigned a trust level, denoted as Trust(a), and an imagination factor ranging from $0<a<1$. Values approaching 0 indicate higher divergence (greater tolerance for uncertainty), whereas values near 1 signify lower tolerance for uncertainty. For each scientific or technical problem, rounds of generation and debate $y(y\geq 2)$ are conducted. For each formulated problem, CSTrainer selects m candidate researcher agents, denoted as Rm (where, $2\leq m\leq M$), from each of the 5 agent clusters to perform ontology planning. The remaining M′ (M'+m=M) agents assume the role of judges, voting to select the most rational ontology components. Following this methodology, the user requirements are defined as concepts, the derived scientific/technical problems as entities, and the adopted technical routes and workflow execution strategies as functions within the ontology. By assigning necessary computational attributes or activity definitions, a distributed dynamic ontology network is constructed, spanning from problem formulation to resolution, as the instance illustrated in fig—2.

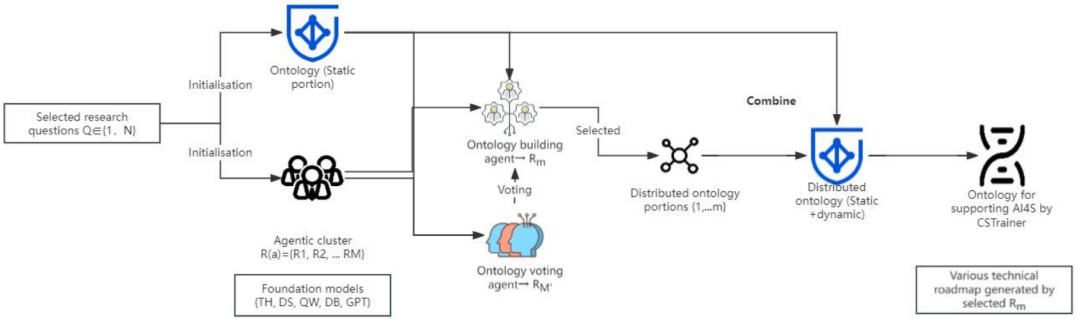


Figure 2—Distributed ontology structure

The selection distribution of Rm follows a near-Gaussian distribution, concentrating core candidates within a mid-to-high confidence interval while distributing peripheral candidates in a gradient manner. This approach ensures fundamental rationality in ontology planning while preserving the capacity for multi-source exploration of technical routes. Each Rm independently plans both the static (entities and classes) and dynamic (computational attributes, functions, or activities) components of the ontology corresponding to the scientific problem. The static components anchor the core entities and intrinsic attributes, whereas the dynamic components leverage the specific confident levels and imagination factors of the agents. This

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provides diversified possibilities for constructing technical routes, task scheduling, workflow execution, model training strategy configurations, result evaluation protocols, and iterative feedback policies. The cognitive biases and capability characteristics of various Rm agents complement one another, covering a multi-dimensional spectrum of ontology construction strategies.

Upon completing the planning of the dynamic ontology components, the M′ judge agents (M′=M−m, and M′must be an odd number) cast votes based on the scientific rigour, feasibility, innovativeness, and alignment with the scientific problem. This screening process selects the more rational ontology construction scheme for the current round. Following each voting round, CSTrainer autonomously adjusts the confidence levels of each Rm via an evolutionary mechanism. This process employs a weighted Bayesian probability evaluation method (Guo et al. 2022), fusing the initial trust level of Rm, the performance weight of the current round's planning, and the quantified results of the judges' votes to achieve dynamic confidence updates. Consequently, agents demonstrating superior performance gain higher confidence, securing a more central role in subsequent ontology planning rounds. Specifically, integrating the initial trust level, current round performance weight, and quantified voting results, the confidence Pi(t+1) of the i-th candidate agent Rmi after a single iteration is constructed using the weighted Bayesian evaluation formula based on the posterior probability P(correct| Vi) of the current round's voting.

$$P_i(t+1) = P_i(t) \cdot \left[\omega \cdot P\left(\text{correct} \middle| V_i\right) + (1-\omega) \cdot \sigma_i\right] \tag{1}$$

where, $P_i(t)$ denotes the initial confidence level of agent R_{mi} (i.e., the trust level prior to the current iteration), and $P_i(t+1)$ represents the updated confidence level after iteration. The parameter $\omega \in [0,1]$ is the weighting coefficient for the Bayesian posterior probability, characterising the importance of the judge agents' voting outcome in the confidence update. The term $(1-\omega)$ serves as the coefficient for σ_i, the performance weight of R_{mi} in the current round of ontology planning. The value of $\sigma_i, \in [0,1]$ is quantified based on dimensions such as the scientific rigour and feasibility of the ontology blueprint, satisfying the normalisation constraint $\omega+(1-\omega)=1$. $P(\text{correct}|V_i)$ is the weighted Bayesian posterior probability, representing the likelihood that R_{mi}'s ontology planning scheme is correct given that it received V_i supporting votes from the judges. This is derived via simplification of Bayes' theorem as follows:

$$P\left(\text{correct} \middle| V_i\right) = \frac{P\left(V_i \middle| \text{correct}\right) P_0}{P\left(V_i \middle| \text{correct}\right) P_0 + P\left(V_i \middle| \text{incorrect}\right)\left(1-P_0\right)} \tag{2}$$

In the above mentioned equation, P_0 represents the prior probability (defaulting to the baseline probability of 0.5 for a correct ontology plan). $P(V_i|\text{correct})$ denotes the probability of receiving V_i supporting votes when the plan is correct, while $P(V_i|\text{incorrect})$ denotes the probability of receiving V_i supporting votes when the plan is wrong. Both probabilities are weighted by the overall trust level of the judge agents $R_{M'}$.

The entire ontology construction and iteration process incorporates a HIL evaluation mechanism, since AI4S process is a in-depth human-machine interaction. Human experts intervene by providing standardised prompt based feedback, issuing a binary evaluation of 0 (Negative) or 1 (Positive) on the ontology segments generated by each R_m. Correspondingly, the trust level R(a) is adjusted by −0.1or +0.1. This binary human feedback is integrated into the weighted Bayesian probability model as critical prior evidence, participating in the adjustment of R_m's confidence level. This approach not only compensates for the limitations of autonomous agent evaluation but also reduces the cognitive burden on human experts through a lightweight feedback format, achieving human-agent collaborative optimisation of the ontology.

Throughout the y rounds of generation and debate iterations, CSTrainer continuously collects the static and dynamic components generated by all R_m agents in each round. Integrating the confidence evolution results, judge voting outcomes, and binary human feedback from every round, the system performs independent ranking and comprehensive filtering of the ontology's constituent parts. These include technical routes, task planning, workflow design, model algorithm training and configuration

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schemes, result evaluation protocols, and feedback iteration strategies. The ranking process centres on the rationality, adaptability, and innovativeness of each component, while assigning higher weights to content generated by higher confidence R_m agents. Finally, the optimal solution for each part is selected from all candidate construction possibilities, integrated, and reconstructed to form an optimal ontology backbone as the blueprint for the specific scientific problem at hand. This blueprint is both scientifically rigorous and feasible, providing a unified and precise guiding framework for subsequent route planning, workflow formulation, and model training configuration. This ensures the overall controllability and efficient implementation of the scientific research cognitive pathway.

Ontology aided specific process modelling

Following the construction of the complete ontology, CSTrainer employs execution agents to plan the problem-solving workflow and the detailed modelling process for scientific research activities by traversing the ontology's network structure. Adopting a hierarchical, distributed architecture, the system intelligently traverses the 5 selected foundational models, matching capabilities to ensure optimal selection. Leveraging the core characteristics of the static entities within the ontology and the prior constraints of the dynamic components, such as technical routes and task plans—the execution agent cluster achieves end-to-end autonomous planning for the entire pipeline. This covers the confirmation of technical routes and workflows, tool matching, solution formulation, and continuous refinery during the problem-solving process. Anchored by the optimal execution path, the system integrates probabilistic modelling, dynamic weight updating, and autonomous evolution mechanisms. Simultaneously, it capitalises on the advantages of hierarchical collaboration and the complementary capabilities of the foundation models, ensuring the scientific rigour, adaptability, and dynamic optimisation of the execution process.

Upon confirmation of the technical route and workflow, the execution agent cluster conducts refined modelling for each task node within the global workflow, adhering to the fixed technical route defined by the ontology network. The detailed workflow modelling process can be summarised in the following steps:

(1) For each sub-task node in the workflow, the execution agent first extracts the node's task features from the ontology (including task type, solution objectives, data characteristics, etc.). Subsequently, it screens for suitable candidate models from a domain specific model library (encompassing AI models, mechanistic models, and multivariate statistical models), as summarised in Tab 3.

Table 3—Algorithm for appropriate candidate model selection

Algorithm 3: Algorithm for candidate model selection
1: **Input:** Task Node Feature F_n, Domain Model Library Mo, Adaptation Threshold τ_S
2: **Output:** Candidate Adapted Model Set Mo_n
3: Procedure:
4: // Step 1: Initialise Candidate Model Set
5: Mo_n = []
6: // Step 2: Iterate Over Domain Model Library
7: For each model mo_p in Mo:
8: // Compute cosine similarity between task feature and model's adaptive feature
9: sim = cosine_similarity(F_n, mo_p.feature)
10: // Step 3: Filter by similarity threshold
11: if sim >= τ_S :
12: Append mo_p to Mo_n
13: // Step 4: Return final candidate set
14: Return Mo_n

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(2) Utilising the ontological constraints and the candidate model set for each task node as inputs, the 5 foundational models (denoted as M={m1, m2, m3, m4, m5}) are engaged to autonomously generate specific work execution methodologies. Each model produces a proposal encompassing model selection, execution strategies, and result processing protocols, culminating in 5 distinct initial candidate solutions, as detailed in Tab 4.

Table 4—Specific research sub-task initialisation

Algorithm 4: Algorithm to form the initialised specific execution work steps

```
1: Input: Foundation Model Set M, Candidate Model Set Mo_n, Ontology Constraint O_constraint
2: Output: Initial Work Execution Method Set S
3: Procedure:
4: // Step 1: Initialise execution method set
5: S = []
6: // Step 2: Iteration over base model set
7: For each foundation model m_x in M:
8:     // Step 3: Generate model selection component
9:     mo_select = m_x.choose_model(Mo_n, O_constraint)
10:    // Step 4: Generate execution strategy component
11:    st_strategy = m_x.make_strategy(mo_select, O_constraint)
12:    // Step 5: Generate result processing component
13:    r_process = m_x.result_strategy(O_constraint)
14:    // Step 6: Form single model execution scheme
15:    S_x = (mo_select, st_strategy, r_process)
16:    // Step 7: Append scheme to execution method set
17:    Append S_x to S
18: // Step 8: Return final execution method set
19: Return S
```

(3) To identify the optimal solution from the initial candidate set, the execution agent implements a cross voting mechanism. The foundational models evaluate each other's proposals, with solution convergence achieved through iterative cycles of “voting, adjudication, and re-planning”. Each model calculates the semantic similarity (quantifying the degree of endorsement) between its own proposal and the others; a supporting vote is cast if the similarity meets or exceeds a predefined threshold (see Tab 5).

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Table 5—Optimal workflow selection

Algorithm 5: Semantic similarity based optimal workflow selection based on cross voting

```
Input: Initial Scheme Set S, Base Model Set M, Similarity Threshold τ_Sim, Weights alpha/beta/gamma
Output: Vote Count V for Each Scheme
Procedure:
// Step 1: Initialise vote count array with length equal to S
V = [0] * len(S)
// Step 2: Iterate each voting model in base model set
For i from 0 to len(M)-1:
    m_x = M[i]
    S_x = S[i] // Scheme generated by the voting model itself
    // Step 3: Iterate all schemes to be voted
    For j from 0 to len(S)-1:
        If i == j:
            Continue // Skip the scheme of the model itself
        S_y = S[j]
    // Step 4: Calculate semantic similarity of three components
        sim_mo = semantic_similarity(S_x[0], S_y[0])
        sim_st = semantic_similarity(S_x[1], S_y[1])
        sim_r = semantic_similarity(S_x[2], S_y[2])
    // Step 5: Compute weighted total similarity
        total_sim = alpha*sim_mo + beta*sim_st + gamma*sim_r
// Step 6: Increment vote if similarity meets threshold
        If total_sim >= τ_Sim:
            V[j] = V[j] + 1
// Step 7: Return vote counts for all schemes
Return V
```

(4) Following the initial voting round, a proposal is directly adopted if it receives 3 or more supporting votes. If all proposals receive fewer than 3 votes, the top x(x<5) ranked proposals are selected. The foundational models are then prompted to fuse and optimise these candidates to regenerate new proposals. This voting process is repeated iteratively until convergence is achieved (see Tab 6).

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Table 6—Iterative selection of execution process

Algorithm 6: Iterative selection of execution process

1: **Input:** Initial Scheme Set (S), Vote Count (V), Foundation Model Set (M), Ontology Constraint (O_{constraint})

2: **Output:** Optimal Execution Scheme (S^*)

3: // Step 1: Initialise the iteration

4: ($S_t \leftarrow S$, $V_t \leftarrow V$)

5: **repeat**

6: // Step 2: Check whether a proposal reaches the adoption threshold

7: **if** \max(V_t)) **then** // the max vote obtained, V_t should less than max iterative round

8: ($S^* \leftarrow$ argmax (S_t, V_t)

9: **return** (S*)

10: **end if**

11: // Step 3: Select the top-ranked proposals

12: ($S_{top} \leftarrow$ {TopX}(S_t, V_t), while x<5)

13: // Step 4: Fuse and optimise the selected proposals

14: **for each** (m_i in M) **do**

15: (S_i{new}$\leftarrow m_i$, optimise (S_{top},$O_{constraint}$))

16: **end for**

17: // Step 5: Regenerate the proposal set

18: ($S_{t+1} \leftarrow$ {S_1{new}, S_2{new}, ...,S_{|M|}^{new}})

19: // Step 6: Re-evaluate the regenerated proposals

20: ($V_{t+1} \leftarrow$ {CrossVote}(S_{t+1}))

21: ($S_t \leftarrow S_{t+1}$)

22: ($V_t \leftarrow V_{t+1}$)

23: **until** max(V_t) // the max vote obtained, V_t should less than max iterative round

24: **return** (S*)

(5) Following the selection of the optimal solution, the execution agent binds it to the ontology defined workflow. For each task node, this generates a refined execution flow encompassing the adapted model, execution strategy, and result processing protocol. These flows are subsequently integrated to form a complete portfolio of refined process models, thereby facilitating the transformation from the global workflow to an implementable execution pipeline. This approach achieves precise alignment between task nodes and domain-specific models through a linear logic of feature matching, model screening, and scheme generation, ensuring the scientific rigour and rationality of the models underpinning each execution step. Furthermore, the iterative convergence mechanism comprising voting, adjudication, and re-planning, effectively mitigates the cognitive biases inherent in any single model through allied decision making among

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multiple foundational models, guaranteeing that the final execution scheme converges towards the optimal solution. Simultaneously, the entire process is governed by ontological constraints, which strictly limit divergence during refined process modelling while significantly enhancing the adaptability of the scheme to the scientific problem solving objectives through complementary model capabilities. Consequently, this method provides reliable technical support for translating global workflows into actionable execution pipelines, enabling CSTrainer to complete a comprehensive problem sorting process efficiently and autonomously under a controlled cognitive framework.

Experimental evaluation

This study utilises one open access dataset and two proprietary datasets to validate the performance of CSTrainer in both scientific problem formulation and resolution. The 3 datasets are: the UNISIM-IV open-access dataset (UNISIM-IV 2026), a proprietary dataset for the re-evaluation of mature wells (logging data) from CNPC, and a proprietary dataset for transnational pipeline scheduling from CNPC. The data specifications and usage are as follows:

(1) UNISIM-IV open-access dataset: Constructed based on a pre-salt light oil carbonate field in Brazil, this dataset contains 2,891 days of daily production data from 6 producers and 7 CO_2 alternating water injection wells. It covers core indicators such as oil production rate (Qo), bottom hole pressure (BHP), water injection rate (WI), and gas injection rate (GI). The data have been standardised and are applicable to scenarios involving reservoir dynamic prediction and inter-well connectivity inference.

(2) Tight gas mature well re-evaluation proprietary dataset: This dataset comprises desensitised data from 75 tight gas wells (with over 30 years of history) among various regions of China. It includes 6 types of logging curves (e.g., acoustic, resistivity, gamma, etc.), interpreted parameters (e.g., porosity), production performance data, and core analysis results (Chen&Zhao et al. 2025). It is suitable for research on logging curve reconstruction, formation classification, potential layer identification, stimulation candidate recommendation, and intelligent log interpretation.

(3) Transnational Pipeline Scheduling Proprietary Dataset: This dataset integrates operational data from 6 cross-border finished natural gas pipelines (across Central Asia and Southeast Asia) from 2018 to 2024. It contains 180 million entries of time-series/tabular data (SCADA monitoring, inventory, weather, PPS, etc.) and over 800 documents (scheduling reports, geopolitical policy documents, etc.). Covering multiple countries and diverse operational constraints, it is intended for research on pipeline throughput optimisation and collaborative maintenance strategy decision-making (Chen&Yang et al. 2025).

Using these 3 datasets, the study verifies CSTrainer's capability in formulating and solving scientific questions, while also evaluating the scalability of the proposed method across upstream and downstream scenarios in the oil and gas industry. At the commencement of the experiments, relevant prior research published in the last 5 years and utilising these 3 datasets was retrieved. These were used as foundational heuristics for CSTrainer. Given the provided materials, CSTrainer was tasked with proposing potential scientific questions and conducting experimental validation for these problems following the procedures outlined previously. Additionally, the authors executed a complete process—from scientific problem discovery to resolution for each dataset as a benchmark. These execution processes were solidified into metadata, instruction sets, and interface invocation methods via skills.

The experiments were triggered by the authors within the CSTrainer system environment. The entire process—from derivative scientific problem discovery to resolution—was executed via skills, encompassing: the rationality of scientific problem formulation; the rationality of technical route planning; the accuracy of ontology construction; workflow generation capability; and the accuracy of detailed computational schemes. The design of the evaluation experiments is as follows:

(1) Scientific Problem Formulation: For each of the 3 selected datasets, CSTrainer was employed to formulate 3, 5, and 10 scientific questions, respectively. The rationality of these formulations was evaluated

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using a combination of subjective and objective measures. The UNISIM-IV dataset was used to evaluate problem formulation and resolution regarding reservoir dynamic prediction and inter-well connectivity inference. The logging interpretation dataset was used to assess problem-solving capabilities in tight gas well log curve reconstruction, micro-layer identification, potential layer recommendation, and stimulation strategy formulation. The transnational long-distance pipeline scheduling dataset was used to evaluate problem formulation and resolution concerning multi-pipeline collaborative scheduling, inlet-outlet flow difference prediction, pipeline operation NPT (Non-productive time) optimisation, and multi-objective collaborative optimisation of gas procurement and sales strategies.

(2) Technical route planning: Given that the risk of hallucinations and reasoning volatility in CSTrainer may increase when the number of scientific questions exceeds 5, the top 5 problems (by score) were selected for technical route planning. The rationality of these routes was evaluated using a combination of subjective and objective measures.

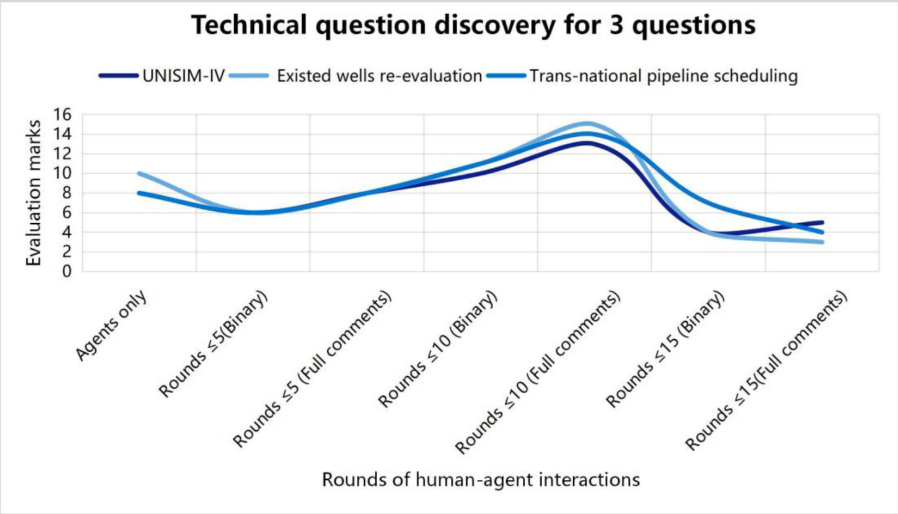
(3) Ontology construction: Based on the planned technical routes, CSTrainer's distributed ontology simulation agent cluster was employed. Through simulation and deduction, the agent cluster autonomously selected different ontology modules to compose distinct ontology architectures tailored to each scientific problem.

(4) Execution planning: Guided by the ontology, execution strategies were generated for the algorithms involved in each module of the technical route, including machine learning algorithms, multivariate statistics, and mechanistic models.

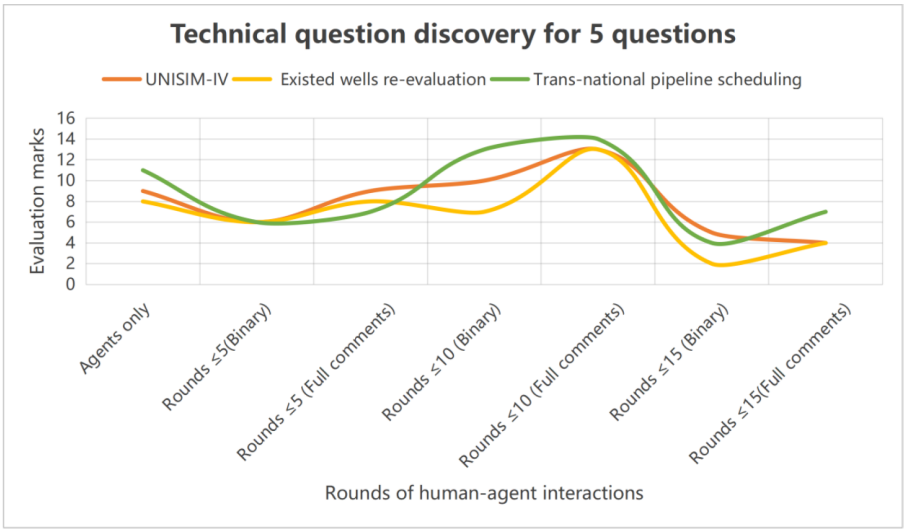
(5) Evaluation metric calculation: A combined subjective-objective approach was adopted. For objective evaluation, the agents interfaced with the 5 different foundational models to conduct cross-voting, where each agent corresponding to a base model cast one vote, accumulated to form the score. For subjective evaluation, a panel of 10 anonymous researchers—each having presided over projects funded by the national natural science foundation of China (NSFC) in relevant fields was recruited. These anonymous referees evaluated the aforementioned work based on their professional expertise, scoring on a scale of -1 (Negative), 0 (Not sure), and 1 (Positive). The final evaluation score was obtained by aggregating the subjective and objective results.

Fig—3 illustrates the performance of CSTrainer in scientific problem formulation. The study evaluates the system across scenarios requiring the extraction of 3, 5, and 10 scientific questions. For human–machine interaction and ablation comparisons, the following settings are considered: (1) a pure agent baseline without any human intervention; and (2) 5, 10, and 15 rounds of human–machine interaction during the CSTrainer reasoning process. Each interaction setting includes two modes: binary feedback (i.e., “correct” or “incorrect” only) and full comments (i.e., feedback including correctness and complete justificatory rationale).

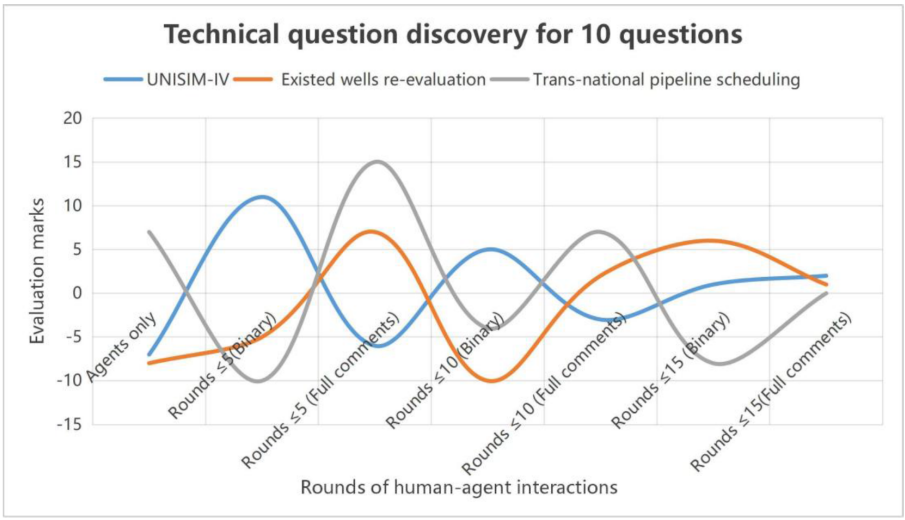
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(a)



(b)



(c)

Figure 3—Technical and scientific question discovered by CSTrainer :
(a) Performance for 3 questions (b) 5 questions (c) 10 questions

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The results in fig—3, based on a multi-scenario experimental design and combined subjective–objective evaluation, systematically characterise the capability boundaries and applicability conditions of CSTrainer in scientific problem formulation. Across three heterogeneous upstream and downstream oil and gas scenarios—reservoir dynamic prediction, re-evaluation of mature tight gas wells, and transnational long-distance pipeline scheduling—CSTrainer reliably extracts 3–5 scientific questions, demonstrating robustness and cross-scenario generalisability. The ablation study further shows that the method effectively captures core contradictions in different scenarios. The relevance and rationality of the extracted problems significantly outperform the pure agent baseline in both subjective and objective evaluations. For instance, under the "3 problems + 10 rounds of full comments" setting, the score improves by 62.5%–87.5% over the baseline, validating its capability for reliable scientific problem discovery and structuring under moderate problem scale and clearly defined scenario boundaries, thereby providing a high-quality starting point for subsequent technical planning.

HIL interaction provides a clear performance gain in scientific problem formulation; however, its effectiveness is jointly constrained by interaction modality and the number of interaction rounds. When extracting 3–5 key scientific questions, full comments interaction yields slightly higher overall scores due to richer semantic and logical guidance. Nevertheless, binary interaction, as a lightweight and cognitively efficient alternative, is sufficient to support effective problem anchoring by CSTrainer and demonstrates basic engineering feasibility. Notably, when the number of interaction rounds exceeds 10, the quality of problem formulation declines markedly under both interaction modes. This suggests that excessive interaction introduces informational noise or leads to objective drift, resulting in distortion effects.

When the number of scientific questions increases to 10 or more, the extraction accuracy of CSTrainer exhibits significant fluctuations across datasets, with subjective–objective evaluation scores ranging from negative values to 15 points. This phenomenon is attributed both to hallucination and instability in handling high-complexity, multi-objective reasoning tasks, and to well-established principles in scientific research practice: high-quality research typically focuses on a limited number of core problems, whereas excessive proliferation of problems dilutes the research focus and exceeds the effective support range of available data and methods. Therefore, the observed performance variability under large-scale problem extraction further underscores the practical value of CSTrainer in assisting researchers to converge on core scientific questions, while also indicating directions for future technical optimisation.

Overall, the experimental results demonstrate that CSTrainer, under a framework characterised by moderate problem scale, focused objectives, human–machine collaborative interaction, and controlled exploratory scope, can effectively support scientific problem formulation across upstream and downstream domains, thereby providing a viable human–AI collaborative paradigm for large-model-assisted scientific discovery.

During the technical route planning phase (fig—4), CSTrainer demonstrated optimal task performance when 5 scientific questions were simultaneously extracted. Accordingly, this study selected the 5 highest scoring core scientific questions from the question refinement stage as inputs to evaluate the system's autonomy and stability in this task. Under the constraint of focusing on these 5 key questions, CSTrainer exhibited markedly enhanced technical route planning capabilities compared with baseline configurations.

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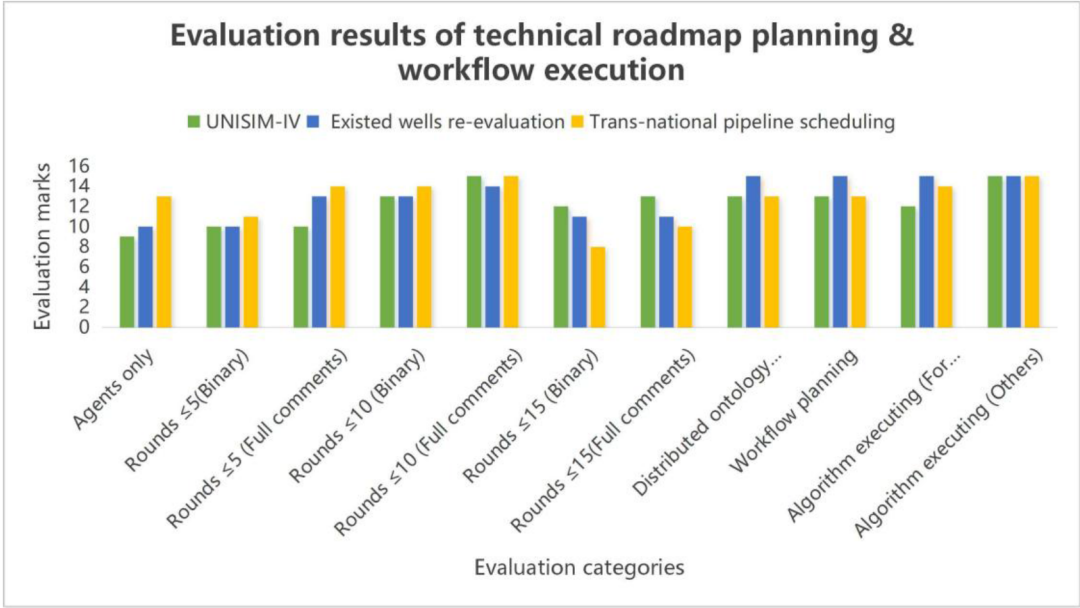


Figure 4—Evaluation results of technical roadmap planning & workflow execution

In the pure autonomous agent mode (agents only), the planning rationality scores for the 3 datasets —UNISIM-IV, tight gas well re-evaluation, and transnational pipeline scheduling reached 9, 10, and 13, respectively, indicating a near self-sufficient reasoning capability. With HIL interaction employed, the scores showed modest improvements. For instance, under the "Rounds ≤10 (full comments)" condition, the composite scores increased to 15, 14, and 15, respectively, corresponding to an average improvement of approximately 15%–20%. However, compared with the scientific question refinement stage, characterised by higher uncertainty, the magnitude of improvement remained relatively limited. This suggests that, in scenarios where problem definitions are explicit, CSTrainer is capable of generating high quality, actionable technical route plans by leveraging its embedded knowledge and planning logic. Its autonomous planning performance approaches, and in some cases matches, that achieved with human guidance. This finding provides empirical evidence supporting the feasibility of enhancing autonomy and reducing reliance on domain experts in AI-assisted scientific research workflows.

The influence of human–machine interaction rounds on planning stability exhibits a quasi-Gaussian distribution. Experimental results indicate that when the number of interaction rounds exceeds 10, no significant fluctuation or systematic degradation in planning performance is observed. Comparing the "Rounds ≤10" and "Rounds ≤15" conditions, score variations across datasets and interaction modes remain within a narrow range of 2–3 points (e.g., UNISIM-IV shows a slight decrease from 15 to 13 under the full comments mode). This behaviour contrasts sharply with the divergence risks observed in the scientific question refinement stage under extended interaction. The difference can be attributed to the more concrete objectives and clearer constraints inherent in the technical route planning phase, which effectively reduce reasoning uncertainty. As a result, the core framework established in earlier interaction stages demonstrates strong robustness. Subsequent interactions yield diminishing marginal improvements but do not lead to substantive degradation in planning quality. Overall, when centred on a limited set of well-defined scientific questions, CSTrainer exhibits high levels of autonomous effectiveness and interaction stability in technical route planning, thereby providing a solid foundation for automated and intelligent scientific research workflows.

As illustrated in fig—4, multi-scenario experimental results demonstrate that CSTrainer achieves a high degree of autonomy and stability across the full pipeline, including ontology representation, workflow orchestration, and algorithm execution. In the distributed ontology construction stage, the system achieved

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strong and consistent performance across three oil and gas scenarios, with scores of 13 (UNISIM-IV), 15 (existing wells re-evaluation), and 13 (transnational pipeline scheduling). Notably, the tight gas scenario involves semantic alignment across heterogeneous data sources such as well logs and core analysis, representing a higher level of modelling complexity. Despite this, CSTrainer achieved a full score, validating its capability to structurally map complex domain knowledge. The identical scores obtained in the UNISIM-IV and pipeline scenarios further indicate robust adaptability across diverse data structures, including time-series production data and multi-source operational documents.

In the workflow planning stage, the score distribution closely mirrors that of ontology construction (UNISIM-IV: 13; tight gas: 15; pipeline: 13), suggesting that high-quality ontologies provide a solid foundation for downstream multi-task orchestration. In particular, within the tight gas scenario, the complex workflow spanning from "well log reconstruction" to "potential layer recommendation" achieved full marks, demonstrating CSTrainer's effectiveness in coordinating multiple modules and orchestrating progressive problem-solving processes. At the algorithm execution stage, the system exhibits strong multi-method integration capabilities. Machine learning based execution achieved scores of 12, 15, and 14, with optimal performance observed in the well defined tight gas scenario. Meanwhile, traditional statistical and physic -based models consistently achieved full scores of 15 across all scenarios. This indicates that CSTrainer not only supports data driven methods but also effectively integrates domain knowledge based models, thereby meeting the demands of complex feasibility analyses in the oil and gas industry.

Figs 5–6 present 2 categories of case studies: (1) future research directions identified by CSTrainer based on user profiles, and (2) end-to-end autonomous processes driven by scientific questions, including ontology inference, data extraction, solution generation, simulation, workflow construction, and execution. These examples demonstrate that CSTrainer can accurately align with users' intentions, prior research outputs, and current research interests to generate targeted and credible future research directions. Furthermore, it can assist researchers throughout the execution of research tasks, significantly improving research efficiency. Moreover, the successful reuse of CSTrainer across 3 upstream and downstream scenarios highlights its strong scalability and reusability as an AI-driven scientific research assistant in the oil and gas domain.

Q: Get me the appropriate resesearch questions and propect technical solution based on prior research attached.
Response by CSTrainer: The problem of improving the scenario adaptability of the self-supervised agent framework across different blocks/well types.
Core Question: The existing framework has been validated to be effective in tight gas existed wells among various blocks, yet it suffers from poor generalisation in model selection and a sharp drop in task execution accuracy when applied to scenarios with distinct geological characteristics, different well types (vertical/horizontal/directional wells), and various reservoir types (conventional oil reservoirs/shale gas/coalbed methane). How to construct a cross-scenario adaptive system for the autonomous genAI agents to achieve stable and high-precision well log interpretation in heterogeneous O&G exploration and development scenarios?

Workflow with Detailed Technical Schemes (Including Models, Training and Modeling Strategies) [Portion]
Step 1: Scenario Investigation and Cross-Scenario Dataset Construction
Technical Scheme: Multi-source oil and gas field data collection and desensitisation processing; standardised data pre-processing for heterogeneous well/reservoir data.
Models/Tools: Data cleaning and normalisation based on Pandas/NumPy; outlier detection via Isolation Forest Algorithm.
Modeling/Training Strategy: Collect well logging, geological, core analysis, and production dynamic data of existed wells from 3-5 typical domestic oil and gas blocks covering different well types and reservoir types; conduct unified desensitisation, missing value interpolation (KNN interpolation), and feature standardisation to construct a labelled cross-scenario validation dataset with block/well type/reservoir type as core tags.
(Ignored step 2..to step 4)
....
Step 5: Knowledge Graph Iteration and Adaptive Strategy Optimisation
Technical Scheme: Supplement and optimise the scenario-model mapping relationship; adjust the fine-tuning parameter strategy of the adaptive module.
Models/Tools: Knowledge graph incremental update (by turtle, could be directly executed by Python); hyperparameter based on Bayesian Optimisation algorithm.
Modelling/Training Strategy: Supplement the scenario-model mapping relationship in the knowledge graph according to the cross-scenario test results, add high-performance algorithm combinations for underperforming scenarios; use Bayesian optimisation to optimize the key hyperparameters (rank, learning rate, batch size) of the LoRA fine-tuning module, and determine the optimal parameter set for different scenario types; retrain the scenario recognition agent with the updated knowledge graph to further improve the accuracy of scenario matching and model selection.

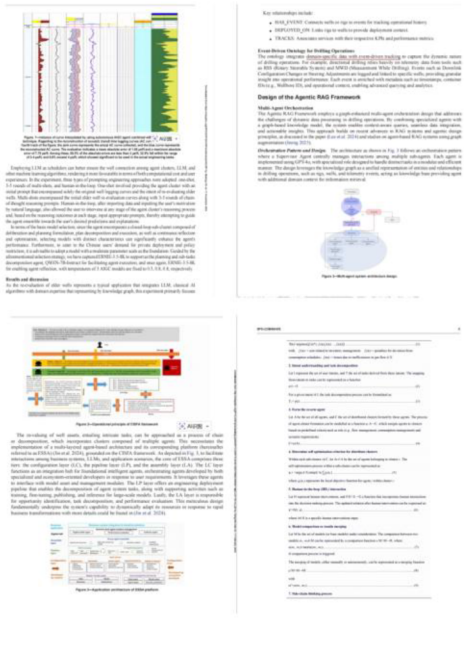


Figure 5—Potential technical selected by CSTrainer (Based on wells re-assessment)

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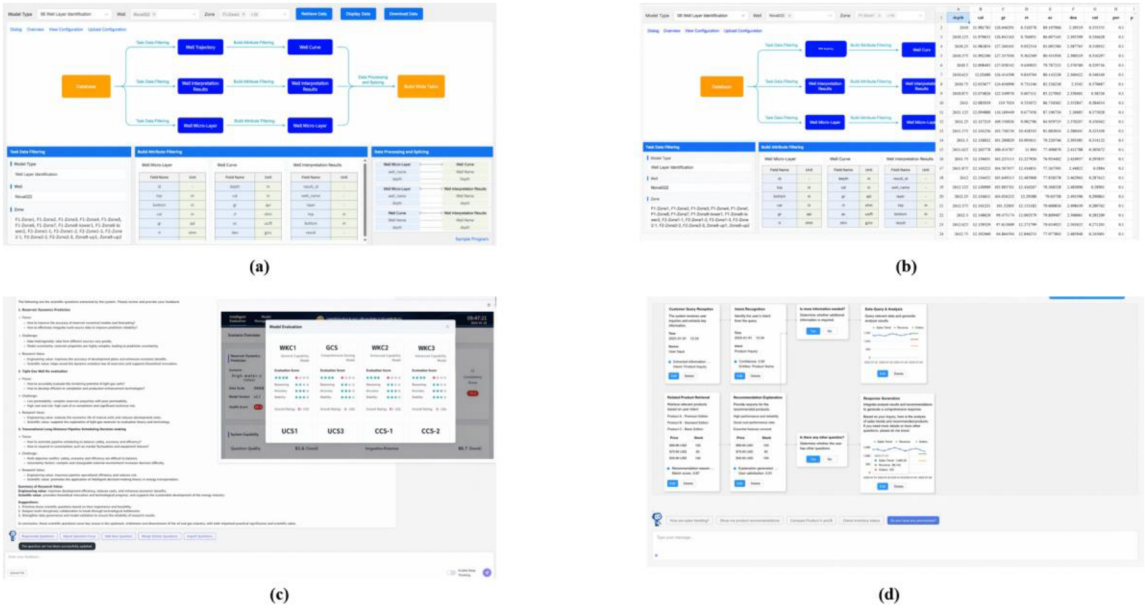


Figure 6—Instance of CSTrainer framework: (a) Well data constructing by CSTrainer's ontology; (b) Appropriate well re-evaluation data set grabbed by; (c) Technical question discovery and simulation (instance) (d) Autonomous workflow generating and executing by CSTrainer

Conclusion and discussion

CSTrainer establishes an autonomous, full-chain research assistance framework encompassing scientific problem formulation - technical route planning - ontology construction - process modelling across 3 distinct upstream and downstream scenarios: reservoir dynamic prediction, mature well re-evaluation, and transnational pipeline scheduling. Its performance is quantitatively and qualitatively validated using multi-source datasets. Quantitatively, during the scientific problem formulation phase, focusing on 3–5 core problems yields a significant improvement in the combined subjective-objective scores compared to agent-only baselines, with the maximum increase exceeding 87%. In the transnational pipeline scheduling scenario, a perfect score of 15 was achieved under the “10-rounds full comments”mode. For technical route planning, agent-only modes scored ≥9 points, with performance further improving by 15%–20% after HIL interaction. In the ontology construction and algorithm execution segments, the tight gas mature well re-evaluation scenario achieved full marks across all dimensions, and the execution of traditional statistical and mechanistic models also received a perfect score of 15. Qualitatively, the system effectively integrates the strengths of multi-model collaboration and human-computer interaction, achieving directional generation of scientific questions remained, structured technical planning, and executable workflow modelling. It demonstrates strong adaptability to heterogeneous data (time-series, logging curves, documents, etc) and various algorithms, maintaining good stability under moderate task scales (3–5 problems). However, volatility increases significantly when the number of problems exceeds 10. This phenomenon demonstrates that CSTrainer can effectively focus on resolving the most significant and relevant research problems. Furthermore, it confirms that the proposed method is suitable for extracting scientific questions that conform to scientific principles and for increasing the proportion of autonomously solved problems. CSTrainer fills the gap in intelligent research assistance tools for the energy and oil & gas sectors, establishing a standardised pipeline from problem discovery to solution implementation. Its multi-agent debate and cross voting mechanisms effectively control the free divergence of foundational models, while the dynamic ontology framework balances scientific rigour with exploratory capacity. The lightweight HIL interaction preserves the researcher's final decision making authority, providing robust technical support for efficient research innovation.

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The system's performance is highly dependent on task scale and interaction design, with the gains from human-computer interaction exhibiting scenario dependent differentiation: Full comments mode outperforms binary mode in problem formulation, whereas in technical route planning, due to explicit constraints, the marginal performance gains from additional interactions diminish. Beyond providing a viable AI-assisted research method, CSTrainer's innovative multi-round, multi-agent cluster debate format offers researchers a means to identify reliable scientific questions and generate technical routes. Furthermore, the results provide a reference for balancing inference length and output timing in real production environments for agent systems in the AI for science domain. It also informs future research on reliable data scales, diversity, noise injection ratios, and quality boundaries, as well as standards for AI-assisted research in the oil & gas sector. Overall, the experimental design and results indicate that the system does not exhibit a strong dependency on specific foundational models. This is attributed to CSTrainer's design, which enables models to objectively respond to research intents through mechanisms such as incorporating user research backgrounds, innovative prompt engineering, continuously evolving search strategies, and multi-model fusion mechanisms. This design enhances system flexibility across different base models, avoids the rigidification of scientific thinking caused by fine-tuning or pre-training, and enables more open and flexible research applications. Moreover, memory based continuous evolution further supports personalised generation based on researcher preferences, providing more effective assistance for scientific exploration.

Regarding limitations, the current CSTrainer primarily focuses on identifying latent scientific questions based on existing researcher foundations and generating corresponding technical routes and execution paths. Its capability boundary remains largely within the realm of planning simulation and software workflow automation, without yet forming a perception-cognition-execution closed loop for real physical environments. Although the system has validated its effectiveness in multiple upstream and downstream oil & gas scenarios, it remains an “offline inference and process orchestration” intelligent assistance paradigm.

Future research will drive the evolution of CSTrainer towards integrating embodied intelligence and world models (Chen&Han et al. 2025). By introducing multimodal perceptual data (e.g., industrial sensors, video streams, control signals) and dynamic mechanistic modelling capabilities, the system aims to transition from static knowledge representation to compute system evolution. Building on this, combined with simulation deduction and feedback optimisation mechanisms, the system is expected to complete strategy verification and risk assessment in virtual environments and further migrate to real production scenarios, form the close loop from technical planning to operational decision optimisation. Simultaneously, at the tool using and process execution level, CSTrainer will evolve from a predefined process-driven mode to an execution framework capable of adaptive tool selection and dynamic strategy optimisation. Coupled with reinforcement learning and planning algorithms, it will achieve continuous self-optimisation in complex task environments. Furthermore, integrating digital twin technology promises to bridge the gap between research design and engineering implementation, constructing a holistic intelligent system spanning problem discovery, solution generation, and operational optimisation. This will facilitate the evolution of CSTrainer from a research assistant tool into an intelligent decision making and execution hub for multiple virtual and physical application scenes within the energy industry.

Conflict of interest

None.

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The authors would also claim the use of AI (deepseek V4 and ChatGPT 4O) for assisting the English clarification, technical words editing, for partial contents of abstract, introduction, related works, system architecture description (including ontology backbone description), conclusion&discussion sections, as well as algorithm proofing. The authors are promised that all AI-assisted parts are thoroughly vetted, fact-checked, and disclosed with published references by the authors, and we hereby confirmed that all AI-assisted portions are 100% obey authors' research intents, and followed the rights of SPE publication policies. Access the latest of CSTrainer wiki from github (will be continuously updated): https://github.com/HChen5965/CSTrainer.

List of symbols and declaration

Symbol	Definition / Description
c	Candidate solution/option in cross-model voting.
C_{rag}	Retrieved domain knowledge/context via RAG.
E	Executor agent within an agentic cluster.
F_n	Task feature vector of a given workflow node.
I	User research intent (background, current prospect, and future directions)
M	Total maximum number of researcher agents in a cluster (should be an odd number)
M_j, m_x	Specific foundation model j or x within the ensemble
M'	Number of judge agents allocated for voting (M' = M - m)
Mo	Domain-specific model library
Mo_n	Candidate adapted model set for node n
N	Target number of scientific questions to be generated/selected
$N_{Candidate}$	Total set or number of candidate scientific questions before final selection
P	Prior research contributions/publications of the user
P_0	Prior probability of an ontology plan being correct (defaulted to 0.5)
$P_{i(t)}$	Initial confidence level of candidate agent R_{mi} at iteration t
$P_{i(t+1)}$	Updated confidence level of candidate agent R_{mi} at iteration t+1
Q*	Final selected set of N scientific research questions
Q_k	Individual scientific research question ($k \in \{1, \ldots, N\}$)
R(a)	Cluster of researcher agents endowed with specific trust and imagination parameters
R_m	Subset of candidate researcher agents selected to conduct ontology planning
$R_{M'}$	Subset of judge agents responsible for evaluating ontology schemes
S	Final confirmed set of research refinement steps or initial work execution methods
S*	Optimal execution scheme selected after cross-model convergence
\α\β\γ	Weighting parameters assigned to individual model execution components during similarity evaluation
σ_i	Performance weight quantifying scientific rigour and feasibility of agent R_{mi}’s scheme
τ_s	Cosine similarity adaptation threshold for domain model selection
τ_{sim}	Semantic similarity threshold required to grant a cross-model endorsement vote
Ω/ω	Weighting coefficient balancing Bayesian posterior probability and performance weight ($\Omega \in [0, 1]$)

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